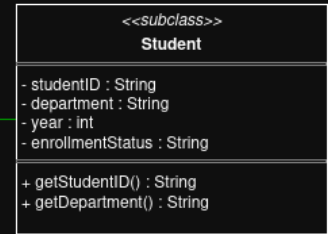
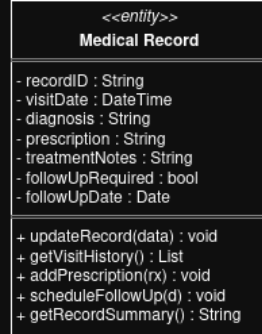
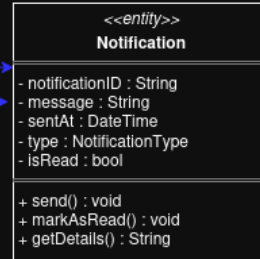
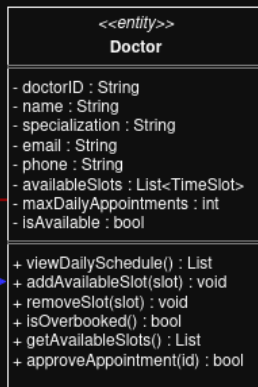
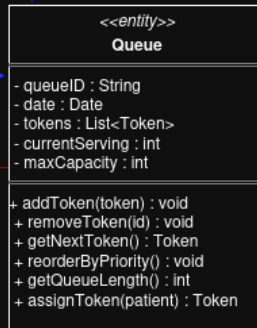
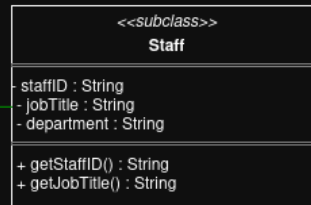
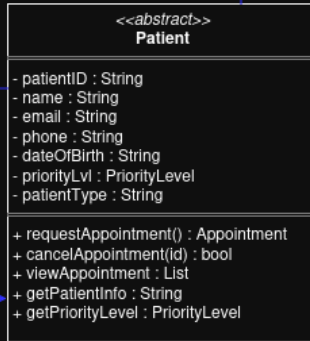
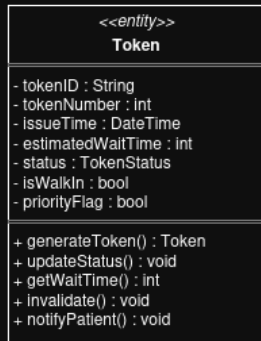
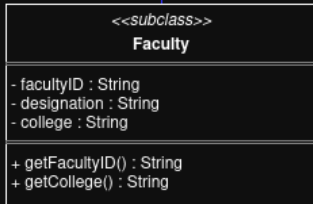
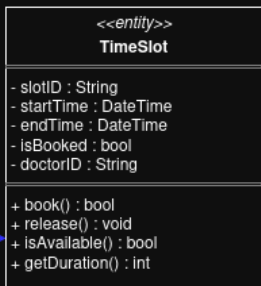
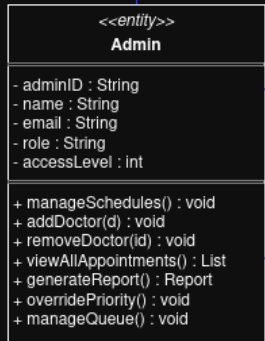
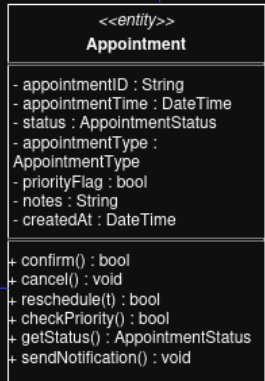


Software Design & Architecture

Assignment #3

Muhammad Aun Anis - 24P-3017



Note: Diagram attached separately for better view, also it was getting too unnecessarily big, unreadable, messy and overall difficult to understand with all the labels etc, so the explanation part is much more in depth!

UML Class Diagram Explanation:

The sections below describe the design decisions captured in the diagram.

1. Identified Classes

Patient (abstract)

- patientID, name, email, phone, dateOfBirth, priorityLevel, patientType
- Abstract base class; cannot be instantiated directly.
- Methods: requestAppointment(), cancelAppointment(), viewAppointments(), getPriorityLevel()

Student, Faculty, Staff (subclasses of Patient)

- Inherit all Patient attributes and methods.
- Student adds: studentID, department, year, enrollmentStatus.
- Faculty adds: facultyID, designation, college.
- Staff adds: staffID, jobTitle, department.

Doctor

- doctorID, name, specialization, availableSlots, maxDailyAppointments, isAvailable
- Methods: viewDailySchedule(), isOverbooked(), getAvailableSlots(), approveAppointment()
- Constraint: isOverbooked() prevents exceeding maxDailyAppointments.

Appointment

- appointmentID, appointmentTime, status, appointmentType, priorityFlag
- status: PENDING, CONFIRMED, CANCELLED, COMPLETED
- Methods: confirm(), cancel(), reschedule(), sendNotification()

Token

- tokenID, tokenNumber, issueTime, estimatedWaitTime, status, isWalkIn, priorityFlag
- Supports walk-in patients via digital token system.
- Methods: generateToken(), updateStatus(), notifyPatient()

Queue

- queueID, date, tokens, currentServing, maxCapacity
- Methods: addToken(), removeToken(), getNextToken(), reorderByPriority(), assignToken()
- reorderByPriority() puts emergency/priority patients first.

MedicalRecord

- recordID, visitDate, diagnosis, prescription, treatmentNotes, followUpRequired
- Methods: updateRecord(), getVisitHistory(), addPrescription(), scheduleFollowUp()

Admin

- adminID, name, role, accessLevel
- Methods: manageSchedule(), addDoctor(), viewAllAppointments(), generateReport(), overridePriority(), manageQueue()

TimeSlot

- slotID, startTime, endTime, isBooked, doctorID
- Methods: book(), release(), isAvailable() — prevents double-booking.

Notification

- notificationID, message, sentAt, type, isRead
- Methods: send(), markAsRead() — triggered by Appointment events.

2. Relationships & Multiplicity

Inheritance

- Patient ← Student, Faculty, Staff (generalization)

Composition

- Patient --* MedicalRecord (1 to 0..*)
- Doctor --* TimeSlot (1 to 1..*)
- Queue --* Token (1 to 1..*) [shown as left→right on same row]

Association

- Patient → Appointment (1 to 0..*)
- Doctor → Appointment (1 to 0..*)
- Appointment → TimeSlot (1 to 1)
- Patient → Token (1 to 0..1)
- Queue → Appointment (1 to 0..*)
- Admin → Doctor (1 to 0..*)
- Admin → Appointment (1 to 0..*)
- Admin → Queue (1 to 1)
- Doctor → MedicalRecord (1 to 0..*)
- Appointment → Notification (1 to 0..*)
- Patient → Notification (1 to 0..*)

3. Advanced Behaviour Constraints

- Overbooking: `Doctor.isOverbooked()` checks confirmed count vs `maxDailyAppointments` before confirming.
- Priority: `Token.priorityFlag` and `Appointment.priorityFlag` mark emergencies; `Queue.reorderByPriority()` sorts accordingly.
- Walk-in: `Token.isWalkIn` distinguishes digital-queue walk-ins from scheduled appointments.
- Double-booking: `TimeSlot.isBooked` is checked atomically inside `Appointment.confirm()`.
- Follow-up: `MedicalRecord.followUpRequired` + `followUpDate` trigger automatic scheduling.
- Access control: `Admin.accessLevel` governs permitted management operations.

4. Enumerations

- `PriorityLevel`: NORMAL, HIGH, EMERGENCY
- `AppointmentStatus`: PENDING, CONFIRMED, CANCELLED, COMPLETED
- `AppointmentType`: SCHEDULED, WALK_IN, FOLLOW_UP, EMERGENCY
- `TokenStatus`: WAITING, SERVING, COMPLETED, CANCELLED
- `NotifType`: APPOINTMENT_CONFIRM, TOKEN_UPDATE, REMINDER, CANCELLATION

